

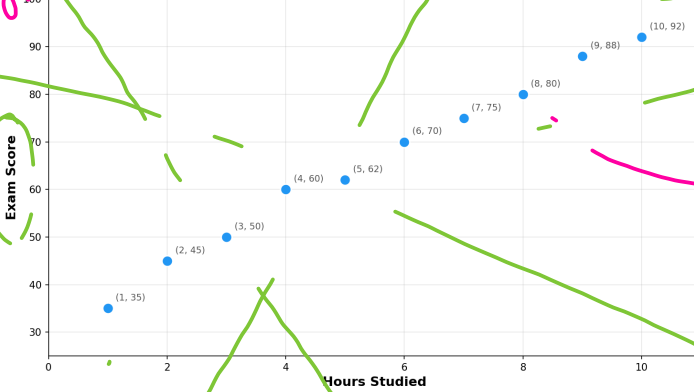


Regression
classification.

Linear.

Student:	1	2	3	4	5	6	7	8	9	10
Hours Studied:	1	2	3	4	5	6	7	8	9	10
Exam Score:	35	45	50	60	62	70	75	80	88	92

10 Students: Hours Studied vs Exam Score



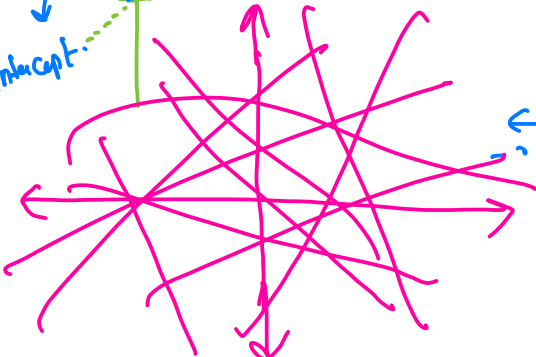
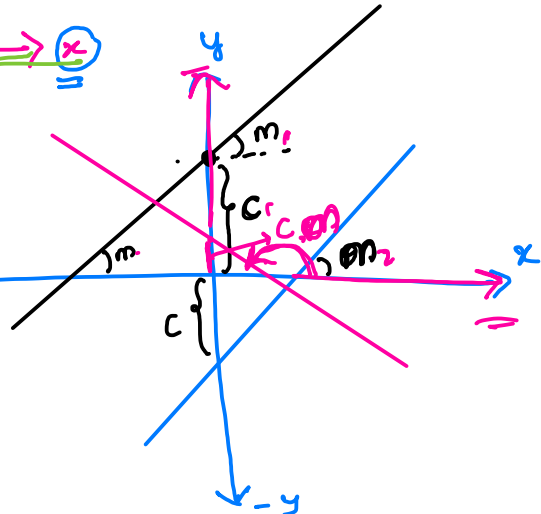
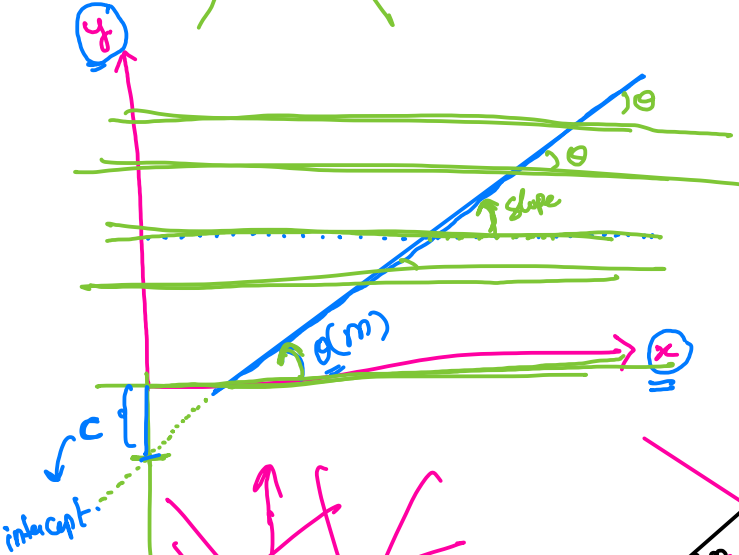
$y = mx + c$

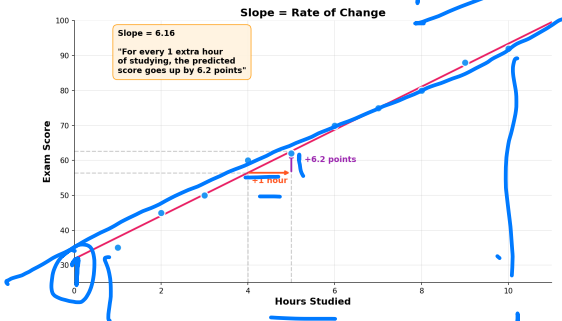
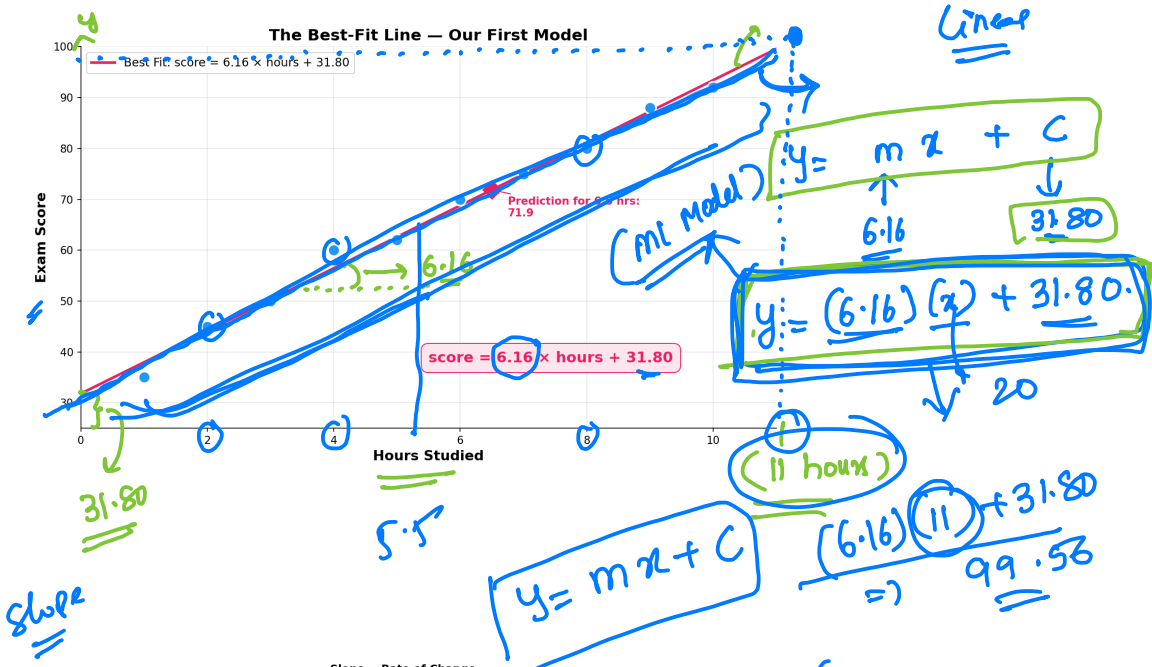
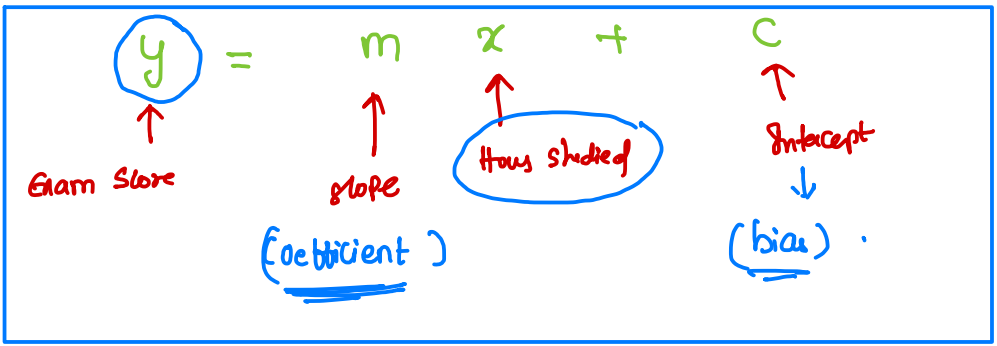
↑ from slope ↑ from studies

equation of a line

$y = mx + c$

↑ slope of the line ↑ intercept





for every 1 hour extra of studying, the predicted score increases by slope value (6.16)

(linear model)
(the rate of change)

$\text{Slope} = 6.16$

$$c = ? \text{ (bias)} \rightarrow 31.80$$

$$y = mx + 31.80$$

$$x = 0$$

$$y = 31.80$$

$$y = mx + c \quad (c = 0) \text{ assume}$$

$$x = 1$$

$$x = 2$$

$$x = 3$$

$$y = m$$

$$y = 2m$$

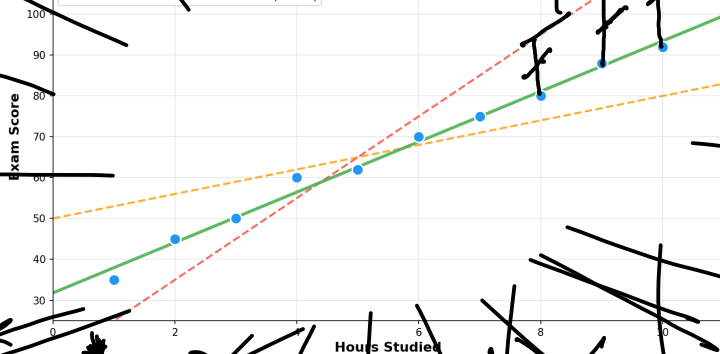
$$y = 3m$$

(slope times)



Which Line Fits Best?

Line A: score = 10 x hours + 15 (too steep)
 Line B: score = 3 x hours + 50 (too flat)
 Line C: score = 6.2 x hours + 31.8 (best fit)



10 diff residual

TR_{L3}

(residual)

(distances)

(as minimum as possible)

Prediction line

$L1$

$L2$

perfect =

TR_{L2}

r_1

r_2

r_3

r_4

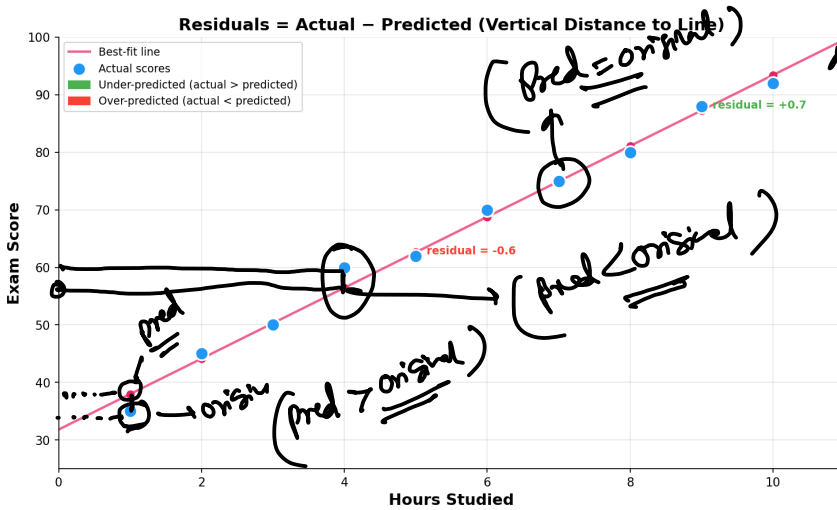
r_5

r_6

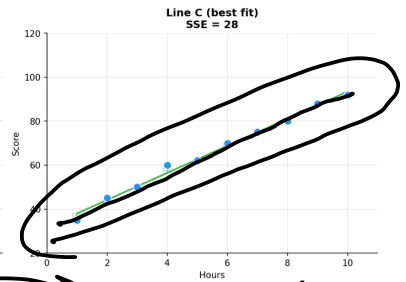
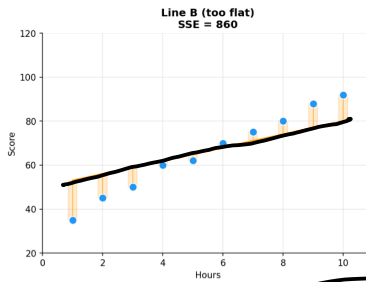
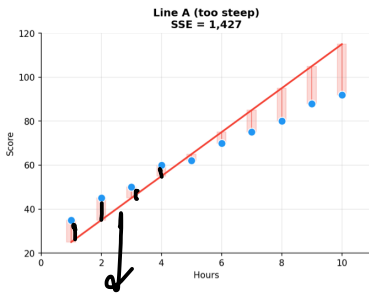
r_7

r_8

total residual diff



OLS: Which Line Has the Smallest Sum of Squared Errors?



m_1, c_1

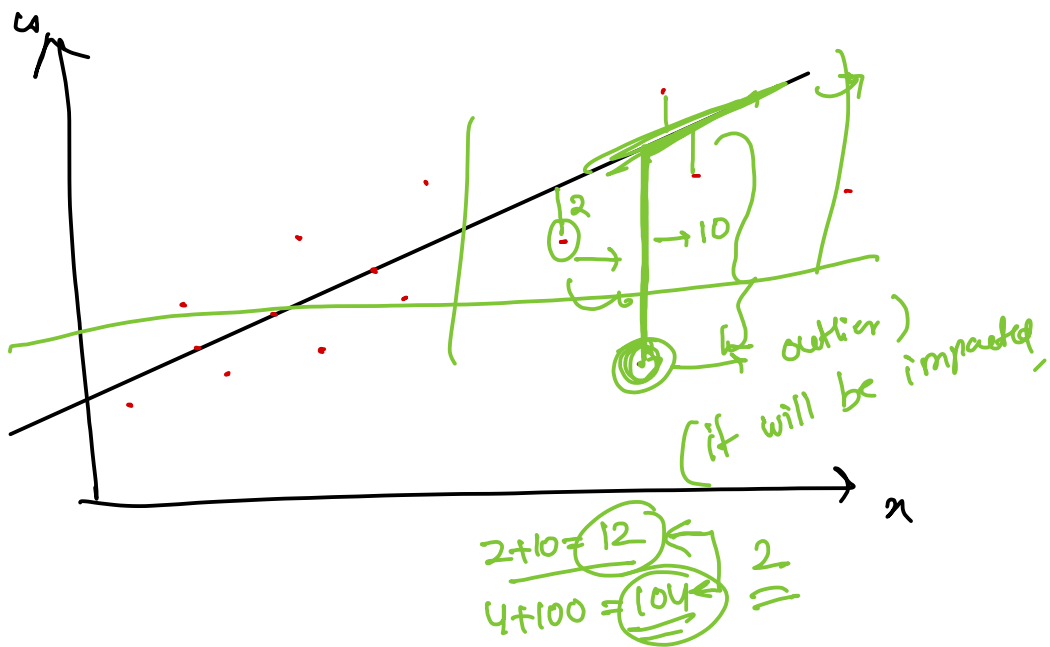
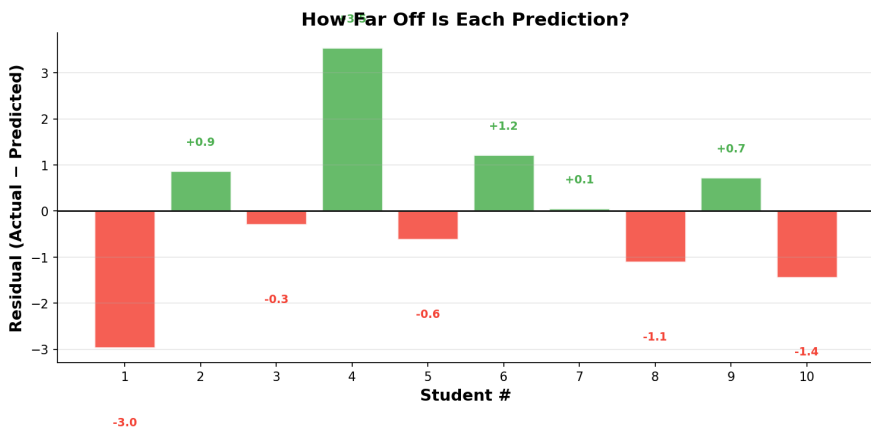
m_2, c_2

m_3, c_3

m_4, c_4

(Linear Reg)

Squash



```
model = LinearRegression()  
model.fit(X_train, y_train)  
  
print(f"Model trained ✓")  
print(f"Number of coefficients: {len(model.coef_)}")  
print(f"Intercept: ₹{model.intercept_:.0f}")  
  
✓ 0.0s  
Model trained ✓  
Number of coefficients: 44  
Intercept: ₹1,324,659
```

